

ECE 3340 Numerical Methods

Homework 5: Numbers in Digital Systems

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Problem 1: Signed Integers

What is the smallest value that can be represented by a 5-bit signed register using two's-complement?

mini: $-2^4 = -16_{10}$

range: -2^{n-1} to $2^{n-1}-1$

mini: $-2^4 = -16$
max: $2^4-1 = 15$

Problem 2: Nintendo Entertainment System Memory

remember there are 246 bytes of addressable memory

A Nintendo Entertainment System (NES) used a pixel processing unit (PPU) with 246 bytes of addressable sprite memory that could be used to process motion on the screen. How many bits were required to address this memory space?

$2^n \geq M \approx M=246 \quad m=b$

$n = 8 \text{ bits}$

$2^n > 246$, we need to find the n bit that is over 246.

$2^8 \rightarrow 256$

$2^7 \rightarrow 128$

$2^6 \rightarrow 64$

n needs to be 8 or higher

Problem 3: NeoGeo Memory

$2^0 = 1024 \rightarrow 2^8 = 2048, 2^9 = 4096, 2^{10} = 8192, 2^{11} = 16384, 2^{12} = 32768, 2^{13} = 65536$

The NeoGeo had 64 kilobytes (kB) of main VRAM. How many bits were required to address this memory space? There is a lot of confusion surrounding the ISO and JEDEC standards for the term kilobyte. Which do you think is used in this case?

VRAM = 64 (kB) $1 \text{ kB} = 1024 \text{ bytes}$

$2^n = n = 16 = 65536$

$64 \times 1024 = 65536$

$2^n = 65536 \text{ bytes} \rightarrow$

JEDEC

Problem 4: Donkey Kong

The number of seconds given to complete each level L in the video game Donkey Kong is $s = \min(10L + 40, 80)$. Why has nobody gotten past level 22?

$s = \min(10L + 40, 80) \approx$ for $L \leq 4: 10L + 40 \leq 80$

$\& L \geq 5: 10L + 40 > 80$ $\leftarrow$ This happens for level 5 and up

Level 22 most likely has an integer overflow causing a kill screen sequence. As the $L \geq 5$, the glitch intensifies making level 22 unbeatable.

Problem 5: Floating Point

Convert the following 8-bit floating point values to decimal assuming the following format (with a bias of 7):

0001 0101 $\rightarrow$

sign exponent mantissa
0 0 0 1 0 1 0 1
3 2 1 0 -1 -2 -3
 $1 + 2^{-1} + 2^{-3} = 0.625$

1.625

$1.m \cdot 2^{e-b} = 1.625 \cdot 2^{-5} = 0.0508$

compute exponents
 $\times 1 - b$
exponents - bias

$2 - 7 = -5$

0.0508